

FOG-ORCHESTRATOR 2.0 -- SIH EVIDENCE PACKAGE

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EVIDENCE PROVENANCE: [AUTHORITATIVE INDEX]

Master navigational directory mapping every SIH presentation claim to verified evidence files with strict provenance classification.

1. Executive Summary & Evaluator Quick Start

In open-cast mining operations (NMDC Bailadila Deposit 5), dense seasonal fog drops sightline visibility below 15 meters, while wet clay slurry reduces friction to $\mu \leq 0.35$. A loaded 165.5-tonne BEML BH100 dumper on an 8-12% downhill ramp requires 35 to 50 meters to stop safely--more than double sight distance. SYNQRA / FOG-ORCHESTRATOR 2.0 provides an end-to-end cyber-physical solution: physical dual-ESP32 prototype, authoritative 3D Digital Twin, 5-constraint physics governor, in-cab driver guidance, and dual-link RF failover.

2. PPT Claim-to-Evidence Matrix

PPT Claim	Evidence File	Verification Method	Provenance
Physical Prototype	03_HARDWARE/.../01_HARDWARE_ARCH	Dual ESP32, optical tachometer, IMU, LoRa	PHYSICAL
Firmware Parity	03_HARDWARE/.../02_FIRMWARE_PARITY	Side-by-side firmware audit & boot IDs	PHYSICAL
Wiring Truth	03_HARDWARE/.../03_WIRING_TRUTH	GPIO pins, SPI bus, H-bridge truth tables	PHYSICAL
V2V/V2I Comm	03_HARDWARE/.../04_COMM_RESULTS	TDM schedule, 11-field V2V, 38.5ms airtime	PHYSICAL (BENCH)
LoRa Safe Beacon	03_HARDWARE/.../05_LORA_GATEWAY	RSSI/SNR, Wi-Fi loss trigger, 1Hz beacon	PHYSICAL (BENCH)
Live Digital Twin	04_DIGITAL_TWIN/01_LIVE_TWIN	Single state store, 10 Hz telemetry sync	SOFTWARE VERIFIED
Predictive Twin	04_DIGITAL_TWIN/02_PREDICTIVE_TWIN	What-if fog lookahead, queueing prediction	PREDICTIVE
NMDC Deposit 5 Twin	04_DIGITAL_TWIN/03_TWIN_RESULTS	Digitized 8-12% ramps, multi-seed determinism	SIMULATION
Physics Governor	05_FOG_GOVERNOR/01_GOVERNOR_ARCH	Stopping sight distance, retarder, friction	MATHEMATICAL
Speed Calibration	05_FOG_GOVERNOR/02_CALIBRATION	Optical staircase, $K_{cal}=34.58$, 42/43 PPR	PHYSICAL (DERIVED)
Fog Speed Response	05_FOG_GOVERNOR/03_GOVERNOR_RESULTS	Safe fog policy, safe crawl/stop behavior	INJECTED + HW LOOP
Driver Assistance	05_FOG_GOVERNOR/04_DRIVER_LOGIC	Cockpit UI, driver in control (no auto-brake)	HUMAN-IN-LOOP
Master Validation	06_RESULTS/01_FINAL_VALIDATION	1,213 backend (100%), 1,860 frontend tests	SOFTWARE VERIFIED
Hardware Bench Traces	06_RESULTS/02_HARDWARE_RESULTS	Motor staircase curves, PWM limits (220/240)	PHYSICAL (BENCH)
Closed-Loop Traces	06_RESULTS/03_FOG_RESPONSE	Weather API injection to safe speed clamp	CLOSED-LOOP
Throughput Retention	06_RESULTS/04_TWIN_RESULTS	68.4% throughput retained in fog vs 0% shutdown	SIMULATION
Research Grounding	07_RESEARCH/01_REFERENCES	DGMS circulars, SAE J1939, ISO 26262, BEML	STANDARDS

3. Evaluator 5-Minute Fast Track

- Step 1: Inspect 01_SYSTEM_OVERVIEW for the complete architecture in 2 pages.
- Step 2: Inspect 03_HARDWARE_AND_COMMUNICATION for physical ESP32 wiring, firmware parity, and RF traces.
- Step 3: Inspect 05_FOG_SPEED_GOVERNOR for the monotonic safe-speed response table.
- Step 4: Inspect 06_RESULTS_AND_VALIDATION for 100% test pass evidence and latency budgets.